

Harish Suresh

San Jose, CA | 812-361-6275 | harishsuresh.contact@gmail.com | linkedin.com/in/harishs163 | github.com/harishs2000

EDUCATION

Master of Science in Computer Science

Indiana University, GPA: 3.93/4.0

Aug. 2024 – May 2026

Bloomington, IN

Bachelor of Engineering in Computer Science

Anna University

Aug. 2018 – May 2022

Chennai, India

EXPERIENCE

Software Engineer

Jun. 2026 – Present

Heartland Community Network — Adtua client engagement

Remote

- Shipped a React/TypeScript and FastAPI campaign builder on a 3–4 engineer team for Adtua, a digital out-of-home advertising marketplace; replaced email handoffs for 15–20 planners booking signage screens, cutting brief turnaround from 2–3 days to same-day across 200 briefs
- Designed FastAPI reservation APIs with idempotency keys and PostgreSQL exclusion constraints, returning the original booking on retries and HTTP 409 for conflicting screen reservations; verified with 25 integration and concurrency tests
- Moved media validation, H.264 transcoding, and thumbnails off the request path into Celery workers (Redis, S3, FFmpeg) with capped backoff retries, terminal failure states, and OpenTelemetry spans across API and worker
- Built a LangChain campaign planner with eligibility enforced in code (budget, location, availability), Qdrant ranking, and an LLM drafting from the ranked set; on a 42-brief pilot all 40 eligible briefs produced valid plans and both ineligible briefs returned no match

Software Developer

Aug. 2022 – Jul. 2024

Sify Technologies

Chennai, India

- Developed and maintained 6 Java/Spring Boot microservices integrating SNMP and NETCONF to poll status and push configuration across 10,000+ production network devices from Cisco, Juniper, and others
- Built a Java ETL pipeline normalizing vendor CLI and NETCONF/YANG data into a Neo4j graph of 10k devices and 50k links, replacing spreadsheet tracing with path and impact APIs for network operations (NOC) engineers; SHA-256 change detection and parallel parsing cut nightly refresh from 45 to 30 min
- Built REST APIs over JPA/Hibernate mappings secured with OAuth 2.0 and Spring Security RBAC; Redis caching with TTL and change-driven invalidation cut average latency from 820 to 310 ms
- Replaced manual artifact copies and restarts with Jenkins pipelines building Docker images to ECR and deploying to EC2 for 12 services (the 6 above plus 6 from a cross-team standardization effort); layer caching and parallel tests cut merge-to-deployment from 25–30 to 20 min

PROJECTS

Warden | Python, FastAPI, GitHub API, OpenTelemetry, Docker, GitHub Actions

- Built a GitHub issue-triage agent (labels, duplicate detection, comments, proposed assignees, escalation) with a hand-written multi-step tool loop, step caps, and tool-failure recovery
- Enforced policy in code at the tool-call layer, not the prompt: comment deletion denied, issue closure and assignment gated on human approval; blocked 11 of 12 unsafe or injected actions

SnapSend | Java, Spring Boot, PostgreSQL, Redis, React, TypeScript, Docker, AWS

- Built a file-sharing service with expiring links, using Redis TTL for expiry checks and AWS S3 for storage

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL, JavaScript

Backend: Spring Boot, JPA/Hibernate, FastAPI, Pydantic, SQLAlchemy, Celery, REST/OpenAPI

Frontend: React, Zustand, Tailwind CSS, HTML/CSS

Data: PostgreSQL, Redis, Neo4j/Cypher, Qdrant

Cloud & Delivery: AWS (EC2, RDS, S3, CloudFront, ECR), Docker, Jenkins, GitHub Actions, CI/CD, OpenTelemetry, Linux, Git

Applied AI: RAG, LangChain, tool-calling agents, vector retrieval, LLM evaluation

Testing & Security: JUnit, Mockito, pytest, Vitest, integration and concurrency testing, OAuth 2.0, JWT, Spring Security